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Reversible Data Encryption Algorithm

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Abstract: — **IDEA is a Block Cipher working on 64 bit Plain Text Block with 128 bits key. Reversible logic implementation of IDEA implements a novel means of reversible cryptic implementation proposed in this paper. An RDEA algorithm with a REV module in the classical IDEA algorithm with 4 bits Plain Text Block and a 6 bits key is proposed. The proposed algorithm is further tested and analyzed for non-linearity in this paper. This algorithm is proposed to improve the complexity and nonlinearity of classical crypto-circuits.**

Key Words: — **Reversible, Reed Muller, Non-linearity, Block cipher.**

I. Introduction

Cryptography is the art & science of achieving security encoding messages to make them non-readable. There are different types of cryptographic algorithms, symmetric key or private key cryptography & asymmetric key or public key cryptography. In symmetric key cryptography a single key is used in both the sender & receiver side to encrypt & decrypt the message, but in a case of asymmetric key cryptography two types of keys are used: public key & private key. The public key is known to all but private key is known either by the sender or the receiver. Such as: ASSYMETRIC KEY: RSA algorithm, Knapsack algorithm. SYMMETRIC KEY: DES algorithm, 3-DES algorithm, IDEA.

The use of quantum mechanical effects to perform various tasks or to break cryptographic systems is described by the Quantum Cryptography [4]. The use of classical (i.e., non-quantum) cryptography to protect against quantum attackers is also often considered as quantum cryptography. Some Quantum Cryptographic gate i.e. C-NOT Gate, C^2-NOT Gate or Toffoli Gate and some ideas of Fredkin Gate, Swap Gate etc have been used in this paper.

Reversible implies a unique output can be achieved by giving a unique input, similarly for this given unique output a unique input can be achieved too. Another way of understanding irreversibility is to think of it in terms of information ensures. If a logic gate is irreversible, then some of the information has been erased by the gate.

Conversely, in a reversible computation, no information is ever erased, because the input can always be recovered from the output.

The International Data Encryption Algorithm (IDEA) [2] is perceived as one of the strongest cryptographic algorithms. Although it is quite strong, IDEA is not as popular as DES and therefore, must be licensed before it can be used in commercial applications. Secondly DES has a long history and track record as compared to IDEA. However one popular email privacy technology known as pretty good privacy (PGP) is based on IDEA.

Therefore in this paper a new algorithm i.e. Reversible Data Encryption Algorithm (RDEA) is introduced from IDEA. It is also a block cipher, it uses both confusion and diffusion technique for encryption, yet its faster than IDEA.

II. .Classical Idea

IDEA [2] is quite strong Cryptographic Algorithm. It is a block cipher. It implies rather than encrypting 1 byte at a time, a block of bytes can be encrypted at a go. In IDEA same algorithm is used for both encryption and decryption. It uses both confusion and diffusion technique for encryption.

This Algorithm [3] works on 64 bit Plain Text Block and the key is longer and consists of 128 bits and it has 8 rounds and 1 Output Transformation. This 64 bit plain text block is divided into 4 parts each size 16 bits. And in each round 6 sub-keys are generated from the original key each consists of 16 bits. These 6 sub keys are applied to the 4 input blocks i.e. P_1 to P_4. In the 1st round as we have six keys K_1 to K_6. So from the 2nd round there will be keys K_7 to K_{12}.Now onwards from the 8th round there will be keys K_{43} to K_{48}.And finally the Output Transformation, consists of 4 sub-keys K_{49} to $K_{52.}$ The ultimate final output comes from Output Transformation. This Output consists of 4 blocks of cipher text i.e. C_1 to C_4 and each of them are 16 bits, after combining them final 64-bit Cipher text block can be achieved.

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14 steps of each round

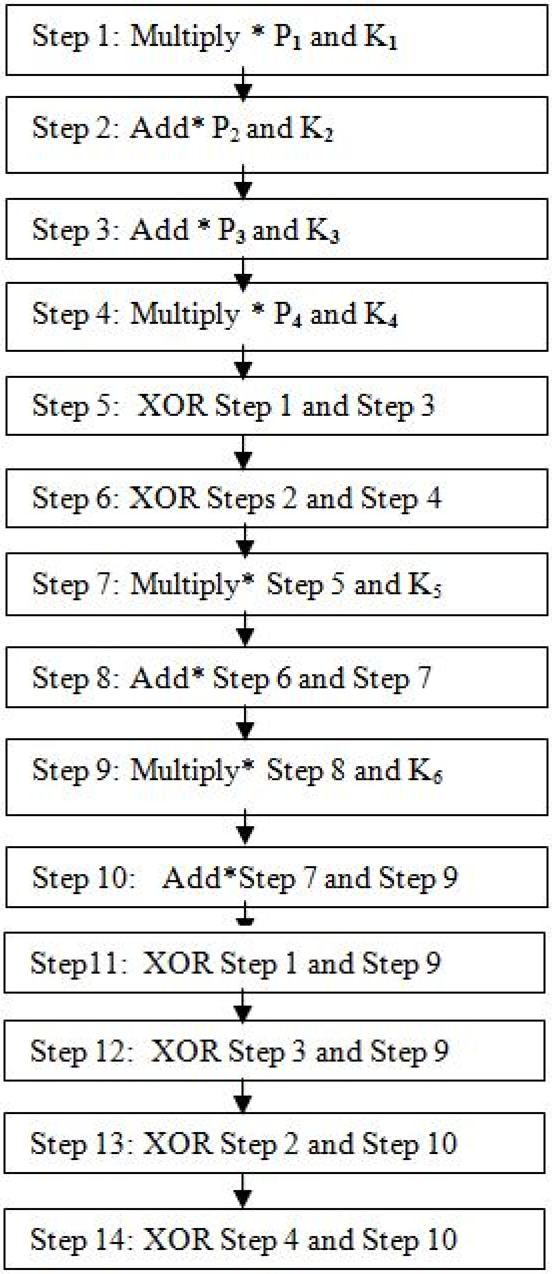

Figure 1. Details of one round in IDEA

In each case of Addition and Multiplication a * mark is used, because in the IDEA algorithm in case of addition the addition modulo 2^{16} is used and in case of multiplication $2^{16}+1$ is used. This modulo operation is used because if two 16 bits have been add or multiply, then the result might be of 17 bits. So to get this unchanged result as the 16 bits no, here this modulo operation has been used.

Let us know the different sub keys which are used in these various steps, from where they have been generated. Initially the key consists of 128 bits. In the 1st round 6*16=96 bits have been used from the 128bits. Now only 128- 96=32 bits have been left. But in the 2nd round 96-32=64 more bits are needed, so for generating these 64 bits a new technique is being applied known as Key Shifting. When all the required bits for a round are not available, then the key 25 bits have to be shifted left circularly. In this key shifting technique the new key is being started with the 26th bit of that old key and being ended with the 25th bit of that old key.

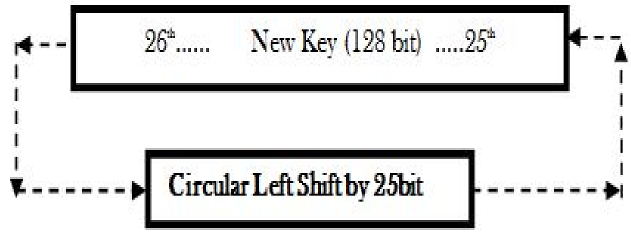

Figure 2. Key Shifting Technique of IDEA

After completing the 8th round there are four outputs, each consists of 16 bits (say R_1 to R_4); will be the input of the Output Transformation and the four sub keys are also needed, each consists of 16 bits (say K_{49} to K_{52}). Now the operations are:

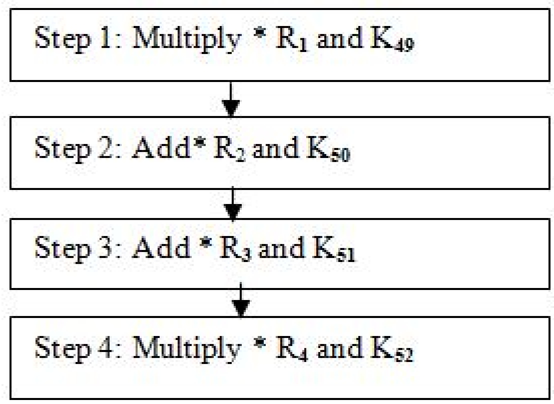

Figure 3. Details of the Output Transformation in IDEA

Sub key generation for this process is same like the previous one and the process of key shifting after 8th round will be continued to generate those sub keys.

III. Reversible Implementation

Let us have a look on a new algorithm i.e. Reversible Data Encryption Algorithm. It is also a quite strong algorithm like IDEA. It is a Block Cipher. It uses both Diffusion and Confusion technique and this algorithm is Reversible. It works on 4 bits Plain text block and 6 bits key and this algorithm has 8 rounds, 1 Output Transformation, 1 Unique Transformation and 1 Reversible Transformation.

A. Working Principle

At first the 4 bit plain text block is divided into 4 parts, each consists of 1 bit (P_1 to P_4) and each 8 round, 6 sub keys will be generated from the original key and will consist of 1 bit i.e. k_1 to k_6. Thus in the 1st round, the plain text P1 to P4 and the sub keys K1 to K6 will be engaged in operations.

B. Sub Key Generation

In the 1st round the 6-bits key have been exhausted. Now again another 6-bits key for the 2nd round have been needed, therefore the Key Shifting technique has to be done. In this

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shifting the previous key has been shifted to 1bit by left circularly; so the new key is being started with the 2nd bit of that old key and being ended with the 1st bit of that old key.

Now here is the diagram:-

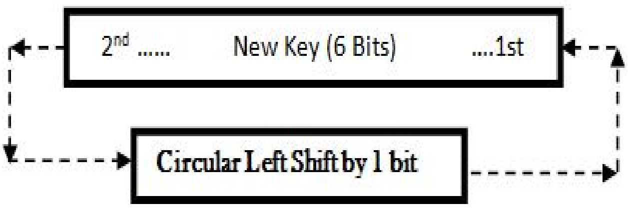

Figure 4. Key Shifting Technique for RDEA

In this algorithm each round is having 14 operations those are:-

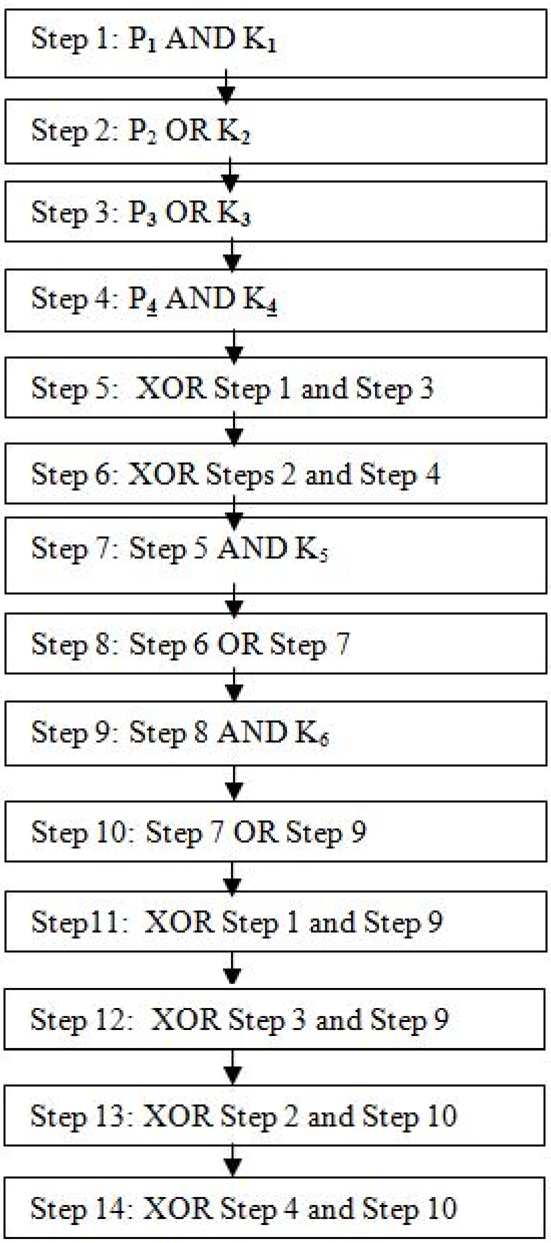

Figure 5. Details of one round in RDEA

Here all the operations which have been used bitwise AND, OR and XOR operations; to maintain the result as preferred bit number.

C. Output Transformation

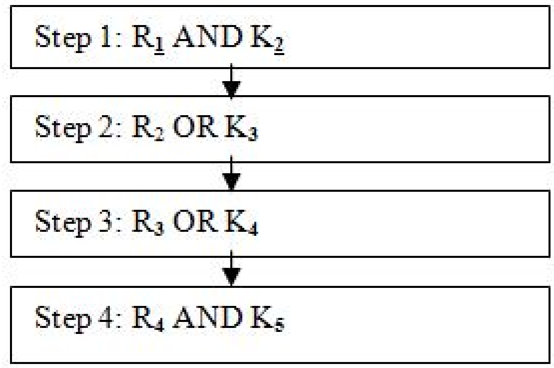

Figure 6. Details of Output Transformation in RDEA

These are the steps of Output Transformation. In this case also all the operations which have been used bitwise AND, OR and XOR operations. Sub key generation for this process is same like all of the 8 rounds. So the sub-keys are K_2 K_3 K_4 K_5

D. *Unique Transformation*

Now the output of Output Transformation which is the input of Unique Transformation is named as *Output-1*.In this Step some Transactions have been made, those are shown in the fig:-

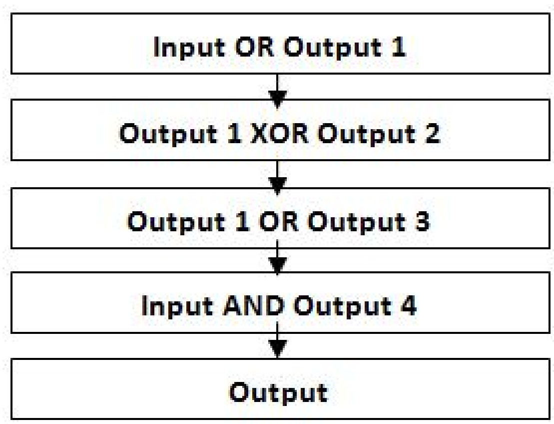

Figure 7. Details of Unique Transformation in RDEA

And finally the output of Unique Transformation is the input of Reversible Transformation.

E. Reversible Transformation

This Transformation is used to make this algorithm Reversible [4] and in this Transformation again these Outputs have been divided into 4 Parts i.e. w, x, y and z and some mathematical operations have been done to achieve the 4 Reversible equations; the 4 Reversible Equations are:-

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1. $1 \oplus yz \oplus wy \oplus wyz \oplus wx \oplus wxy$
2. $1 \oplus w \oplus xy \oplus xz \oplus yz \oplus wxz \oplus wyz \oplus xyz$
3. $1 \oplus w \oplus y \oplus z \oplus yz \oplus wx \oplus wy \oplus wxy \oplus wxz$
4. $1 \oplus w \oplus z \oplus wy \oplus wz \oplus xz \oplus xy \oplus wxz \oplus xyz$

IV. Circuit Diagram

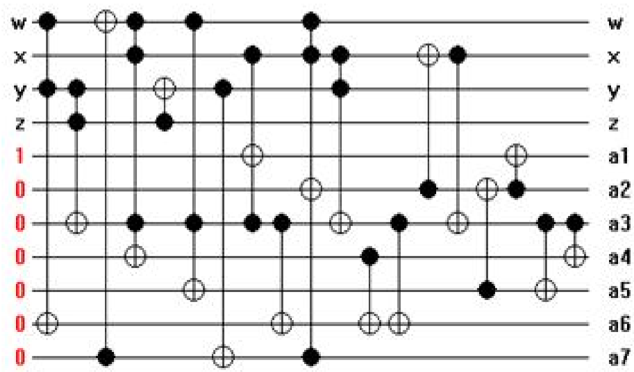

Figure 8. Circuit Diagram of RDEA

V. Matrix Representation

This Matrix is originally $2^{11} * 2^{11}$ Matrix. But here 16 inputs & 16 outputs is our only concern. So the Output Table is:

Table I. Matrix Representation of RDEA

INPUT	OUTPUT
00001000000	00000010000
00011000000	00000010001
00101000000	00111010010
00111000000	00111010011
01001000000	10000010100
01011000000	11000010101
01101000000	11000010110
01111000000	10111100101
10001000000	10000011100
10011000000	10000011101
10101000000	11000011110
10111000000	11111101101
11001000000	00111011000
11011000000	01011101001
11101000000	00000001010
11111000000	00110111011

The Reed Muller [4] equations are Reversible in nature.

The result is represented by matrix of size $2^{11}*2^{11}$. There is only 16 values of concern. This representation is a pars representation where M is the all values of that matrix and corresponding all values are 0 and R_M and C_M are the corresponding values of rows and columns respectively.

M = [1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]

R_M = [64 192 320 448 576 704 832 960 1088 1216 1344 1472 1600 1728 1856 1984]

C_M = [16 17 466 467 1044 1557 1558 1509 1052 1053 1566 2029 472 745 10 443]

VI. Result

The Non-linearity [4] of the proposed RDEA algorithm is

$$\textbf{Dist}\,(T_1, T_2) = \textbf{wt}\,(T_1 \oplus T_2)$$

$$\textbf{nl (RDEA) = 7.}$$

References

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